# TabuLLM: AI Benchmark Report

## 1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

| Audit Indicator | Global Session Mean (± Std) |
|---|---|
| Synthetic Fidelity (JSD) | 65.1 ± 5.07 |
| Correlation Stability | 85.1 ± 3.11 |
| Distribution Consistency | 74.1 ± 5.37 |
| Entropy Preservation | 79.6 ± 2.87 |
| Mean Z Score Drift | 0.3 ± 0.06 |
| Categorical Fidelity | 0.0 ± 0.00 |
| Privacy Risk | 39.4 ± 8.36 |
| Data Density | 100.0 ± 0.00 |
| Coverage Score | 53.2 ± 10.91 |
| Feature Accuracy | 86.0 ± 3.66 |
| Novelty Score | 0.5 ± 0.10 |

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## Dataset Audit: parkinsons.csv

### 1. Dataset Portrait

| Attribute | Value |
|---|---|
| Rows | 195 |
| Columns | 23 |
| Numeric features | 23 |
| Categorical features | 0 |
| Missing cells (%) | 0.0% |

### 2. Generation Strategy & LLM Config [1]

| Parameter | Configuration |
|---|---|
| Algorithm (*12) | gpt-4.1-nano |
| Folds (*13) | 10 |
| Context % (*14) | 20% |
| Generation % (*15) | 100% |
| Join context (*16) | True |

### 3. Statistical Significance & Data Drift

| Feature | Orig Mean | Aug Mean | Shift | P-Value | Status |
|---|---|---|---|---|---|
| MDVP:Fo | 154.229 | 158.839 | +4.610 | 3.0% | OK |
| MDVP:Fhi | 197.105 | 189.615 | -7.490 | 3.8% | OK |
| MDVP:Flo | 116.325 | 121.871 | +5.546 | 4.8% | OK |
| MDVP:Jitter | 0.006 | 0.005 | -0.001 | 20.8% | Warn |
| MDVP:Jitter.1 | 0.006 | 0.005 | -0.001 | 21.0% | Warn |
| MDVP:RAP | 0.003 | 0.003 | -0.001 | 19.0% | Warn |
| MDVP:PPQ | 0.003 | 0.003 | -0.001 | 16.2% | Warn |
| Jitter:DDP | 0.010 | 0.008 | -0.002 | 17.2% | Warn |
| MDVP:Shimmer | 0.030 | 0.023 | -0.007 | 24.1% | Warn |
| MDVP:Shimmer.1 | 0.030 | 0.023 | -0.007 | 24.3% | Warn |
| Shimmer:APQ3 | 0.016 | 0.012 | -0.004 | 23.4% | Warn |
| Shimmer:APQ5 | 0.018 | 0.014 | -0.004 | 22.1% | Warn |

### 4. Quality & Fidelity Metrics (Mean ± Std)

| Metric Category | Aggregated Score |
|---|---|
| Synthetic Fidelity (JSD) (*1) | 65.1 ± 5.07 |
| Correlation Stability (*2) | 85.1 ± 3.11 |
| Distribution Consistency (*3) | 74.1 ± 5.37 |
| Entropy Preservation (*4) | 79.6 ± 2.87 |
| Mean Z Score Drift (*5) | 0.3 ± 0.06 |
| Categorical Fidelity (*6) | 0.0 ± 0.00 |
| Privacy Risk (*7) | 39.4 ± 8.36 |

| | |
|---|---|
| Data Density (*8) | 100.0 ± 0.00 |
| Coverage Score (*9) | 53.2 ± 10.91 |
| Feature Accuracy (*10) | 86.0 ± 3.66 |
| Novelty Score (*11) | 0.5 ± 0.10 |

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

*[6] Synthetic Fidelity (JSD): Overall distribution similarity.*
*[7] Correlation Stability: Multivariate relationships preserved.*
*[8] Distribution Consistency: Median feature similarity.*
*[9] Entropy Preservation: Information diversity maintained.*
*[10] Mean Z Score Drift: Standardized distribution shift.*
*[11] Categorical Fidelity: Category distribution similarity.*
*[12] Privacy Risk: Synthetic rows too close to originals.*
*[13] Data Density: Non-missing cells in augmented data.*
*[14] Coverage Score: Synthetic coverage of original space.*
*[15] Feature Accuracy: Mean feature similarity.*
*[16] Novelty Score: Distance ratio to nearest neighbors.*

## 6. Downstream ML Utility: Comparative Analysis

### Model: KNN

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8358974358974359 | 0.7521367521367521 | 0.7974358974358975 | -0.03846153846153846 | |
| F1 | 0.8946275603661134 | 0.8508384218830078 | 0.8734363748550109 | -0.02119118551110243 | |

### Model: DT

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8512820512820515 | 0.7578347578347578 | 0.8769230769230776 | 0.02564102564102561 | |
| F1 | 0.8965052369401196 | 0.83389597619573 | 0.9150243180470279 | 0.01851908110690805 | |

### Model: GB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9205128205128206 | 0.772079772079772 | 0.9282051282051282 | 0.007692307692307675 | |
| F1 | 0.9457690315111311 | 0.8458861786820009 | 0.9524825824720246 | 0.006713550960893466 | |

### Model: SVM

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8102564102564103 | 0.7521367521367521 | 0.7794871794871795 | -0.03076923076923076 | |
| F1 | 0.8848220246893213 | 0.8516293128378333 | 0.8685085522316107 | -0.01631347245771052 | |

### Model: MLP

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7692307692307694 | 0.717948717948718 | 0.7743589743589744 | 0.005128205128205121 | |
| F1 | 0.8599261354078369 | 0.8193853812711526 | 0.8623842108918243 | 0.002458075483987631 8 | |

### Model: LR

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| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.8692307692307694 | 0.7606837606837606 | 0.8358974358974359 | -0.03333333333333334 | |
| F1 | 0.9130547168372463 | 0.8398175791853952 | 0.8951248892191928 | -0.017929827618053572 | |

### Model: XGB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9333333333333333 | 0.772079772079772 | 0.9333333333333333 | 2.2204460492503132e-17 | |
| F1 | 0.9560674056986365 | 0.8524178831577252 | 0.9559761285512682 | -9.127714736830716e-05 | |

### Model: RF

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9102564102564104 | 0.7863247863247864 | 0.9051282051282051 | -0.005128205128205154 | |
| F1 | 0.9402126269306388 | 0.8599559538840092 | 0.9375639554733344 | -0.0026486714573043524 | |

### Model: AB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.9128205128205129 | 0.8034188034188035 | 0.9102564102564104 | -0.0025641025641025 5 | |
| F1 | 0.941802264596966 | 0.87196479689991 | 0.9404009049307185 | -0.0014013596662476036 | |

### Model: NB

| Metric | Real (Baseline) | Generated | Augmented | Delta (Lift) | Sig. |
|---|---|---|---|---|---|
| ACCURACY | 0.7153846153846153 | 0.7293447293447294 | 0.7051282051282051 | -0.010256410256410243 | |
| F1 | 0.765535606787701 | 0.7896774839184053 | 0.7565226641866498 | -0.009012942601051288 | |

## Appendix: Benchmark Data Trace

*Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline*

### A. Baseline: Original Dataset (Full Record Sample)

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 119.992 | 157.302 | 74.997 | 0.00784 | 0.00784 | 0.0037 | 0.00554 | 0.01109 | 0.04374 | 0.04374 |
| 2 | 122.4 | 148.65 | 113.819 | 0.00968 | 0.00968 | 0.00465 | 0.00696 | 0.01394 | 0.06134 | 0.06134 |
| 3 | 116.682 | 131.111 | 111.555 | 0.0105 | 0.0105 | 0.00544 | 0.00781 | 0.01633 | 0.05233 | 0.05233 |
| 4 | 116.676 | 137.871 | 111.366 | 0.00997 | 0.00997 | 0.00502 | 0.00698 | 0.01505 | 0.05492 | 0.05492 |
| 5 | 116.014 | 141.781 | 110.655 | 0.01284 | 0.01284 | 0.00655 | 0.00908 | 0.01966 | 0.06425 | 0.06425 |
| 6 | 120.552 | 131.162 | 113.787 | 0.00968 | 0.00968 | 0.00463 | 0.0075 | 0.01388 | 0.04701 | 0.04701 |
| 7 | 120.267 | 137.244 | 114.82 | 0.00333 | 0.00333 | 0.00155 | 0.00202 | 0.00466 | 0.01608 | 0.01608 |
| 8 | 107.332 | 113.84 | 104.315 | 0.0029 | 0.0029 | 0.00144 | 0.00182 | 0.00431 | 0.01567 | 0.01567 |
| 9 | 95.73 | 132.068 | 91.754 | 0.00551 | 0.00551 | 0.00293 | 0.00332 | 0.0088 | 0.02093 | 0.02093 |
| 10 | 95.056 | 120.103 | 91.226 | 0.00532 | 0.00532 | 0.00268 | 0.00332 | 0.00803 | 0.02838 | 0.02838 |
| 11 | 88.333 | 112.24 | 84.072 | 0.00505 | 0.00505 | 0.00254 | 0.0033 | 0.00763 | 0.02143 | 0.02143 |
| 12 | 91.904 | 115.871 | 86.292 | 0.0054 | 0.0054 | 0.00281 | 0.00336 | 0.00844 | 0.02752 | 0.02752 |
| 13 | 136.926 | 159.866 | 131.276 | 0.00293 | 0.00293 | 0.00118 | 0.00153 | 0.00355 | 0.01259 | 0.01259 |
| 14 | 139.173 | 179.139 | 76.556 | 0.0039 | 0.0039 | 0.00165 | 0.00208 | 0.00496 | 0.01642 | 0.01642 |
| 15 | 152.845 | 163.305 | 75.836 | 0.00294 | 0.00294 | 0.00121 | 0.00149 | 0.00364 | 0.01828 | 0.01828 |
| 16 | 142.167 | 217.455 | 83.159 | 0.00369 | 0.00369 | 0.00157 | 0.00203 | 0.00471 | 0.01503 | 0.01503 |
| 17 | 144.188 | 349.259 | 82.764 | 0.00544 | 0.00544 | 0.00211 | 0.00292 | 0.00632 | 0.02047 | 0.02047 |
| 18 | 168.778 | 232.181 | 75.603 | 0.00718 | 0.00718 | 0.00284 | 0.00387 | 0.00853 | 0.03327 | 0.03327 |
| 19 | 153.046 | 175.829 | 68.623 | 0.00742 | 0.00742 | 0.00364 | 0.00432 | 0.01092 | 0.05517 | 0.05517 |
| 20 | 156.405 | 189.398 | 142.822 | 0.00768 | 0.00768 | 0.00372 | 0.00399 | 0.01116 | 0.03995 | 0.03995 |

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## B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 136.969 | 166.607 | 66.004 | 0.00923 | 0.00923 | 0.00507 | 0.00463 | 0.0152 | 0.03111 | 0.03111 |
| 2 | 163.656 | 200.841 | 76.779 | 0.00742 | 0.00742 | 0.0038 | 0.0039 | 0.0114 | 0.01659 | 0.01659 |
| 3 | 104.4 | 206.002 | 77.968 | 0.00633 | 0.00633 | 0.00316 | 0.00375 | 0.00948 | 0.03767 | 0.03767 |
| 4 | 125.036 | 143.946 | 116.187 | 0.0128 | 0.0128 | 0.00743 | 0.00623 | 0.02228 | 0.03886 | 0.03886 |
| 5 | 113.166 | 130.27 | 100.673 | 0.00502 | 0.00502 | 0.00257 | 0.00312 | 0.00772 | 0.05279 | 0.05279 |
| 6 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 7 | 135.4 | 165.2 | 65.8 | 0.0091 | 0.0092 | 0.005 | 0.0046 | 0.015 | 0.031 | 0.0312 |
| 8 | 162.8 | 200.0 | 77.0 | 0.0074 | 0.0074 | 0.0038 | 0.0039 | 0.0114 | 0.0166 | 0.0166 |
| 9 | 104.5 | 206.1 | 78.0 | 0.0063 | 0.0063 | 0.0032 | 0.0038 | 0.0095 | 0.0377 | 0.0377 |
| 10 | 125.0 | 144.0 | 116.2 | 0.0128 | 0.0128 | 0.0074 | 0.0062 | 0.0223 | 0.0389 | 0.0389 |
| 11 | 113.2 | 130.3 | 100.7 | 0.005 | 0.005 | 0.0026 | 0.0031 | 0.0077 | 0.0528 | 0.0528 |
| 12 | 157.8 | 173.0 | 68.4 | 0.0036 | 0.0036 | 0.002 | 0.002 | 0.0059 | 0.0372 | 0.0372 |
| 13 | 110.417 | 131.067 | 103.37 | 0.00784 | 0.00784 | 0.00352 | 0.00514 | 0.01056 | 0.03715 | 0.03715 |
| 14 | 208.083 | 253.792 | 91.802 | 0.00757 | 0.00757 | 0.00428 | 0.00428 | 0.01285 | 0.06725 | 0.06725 |
| 15 | 244.99 | 272.21 | 239.17 | 0.00451 | 0.00451 | 0.00279 | 0.00237 | 0.00837 | 0.01897 | 0.01897 |
| 16 | 228.969 | 239.541 | 113.201 | 0.00238 | 0.00238 | 0.00136 | 0.0014 | 0.00408 | 0.01745 | 0.01745 |

## B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 193.03 | 208.9 | 80.297 | 0.00766 | 0.00766 | 0.0045 | 0.00389 | 0.01351 | 0.03044 | 0.03044 |
| 2 | 151.884 | 157.765 | 133.751 | 0.00258 | 0.00258 | 0.00115 | 0.00122 | 0.00346 | 0.0145 | 0.0145 |
| 3 | 154.003 | 160.267 | 128.621 | 0.00183 | 0.00183 | 0.00076 | 0.001 | 0.00229 | 0.0103 | 0.0103 |
| 4 | 117.004 | 144.466 | 99.923 | 0.00353 | 0.00353 | 0.00176 | 0.00218 | 0.00528 | 0.01657 | 0.01657 |
| 5 | 192.818 | 224.429 | 168.793 | 0.03107 | 0.03107 | 0.018 | 0.01958 | 0.05401 | 0.11908 | 0.11908 |
| 6 | 110.707 | 122.611 | 105.007 | 0.00516 | 0.00516 | 0.00277 | 0.00289 | 0.00831 | 0.02215 | 0.02215 |
| 7 | 180.45 | 210.3 | 75.2 | 0.0068 | 0.0068 | 0.0042 | 0.0037 | 0.0125 | 0.028 | 0.028 |
| 8 | 155.2 | 165.8 | 130.4 | 0.0029 | 0.0029 | 0.0012 | 0.0013 | 0.0034 | 0.012 | 0.012 |
| 9 | 160.7 | 170.5 | 125.3 | 0.0018 | 0.0018 | 0.0008 | 0.001 | 0.0023 | 0.01 | 0.01 |
| 10 | 185.3 | 215.7 | 78.5 | 0.0072 | 0.0072 | 0.0048 | 0.004 | 0.013 | 0.029 | 0.029 |
| 11 | 145.8 | 155.2 | 138.9 | 0.0031 | 0.0031 | 0.0014 | 0.0015 | 0.004 | 0.011 | 0.011 |
| 12 | 170.1 | 185.2 | 90.3 | 0.0055 | 0.0055 | 0.003 | 0.0028 | 0.009 | 0.022 | 0.022 |
| 13 | 245.51 | 262.09 | 231.848 | 0.00235 | 0.00235 | 0.00127 | 0.00148 | 0.0038 | 0.01608 | 0.01608 |
| 14 | 115.38 | 123.109 | 108.634 | 0.00332 | 0.00332 | 0.0016 | 0.00199 | 0.0048 | 0.01503 | 0.01503 |
| 15 | 95.056 | 120.103 | 91.226 | 0.00532 | 0.00532 | 0.00268 | 0.00332 | 0.00803 | 0.02838 | 0.02838 |
| 16 | 143.533 | 162.215 | 65.809 | 0.01101 | 0.01101 | 0.00647 | 0.00467 | 0.01941 | 0.05384 | 0.05384 |

## B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 166.605 | 206.008 | 78.032 | 0.00742 | 0.00742 | 0.00387 | 0.00453 | 0.01161 | 0.0664 | 0.0664 |
| 2 | 112.239 | 126.609 | 104.095 | 0.00472 | 0.00472 | 0.00238 | 0.0029 | 0.00715 | 0.05643 | 0.05643 |
| 3 | 148.462 | 161.078 | 141.998 | 0.00397 | 0.00397 | 0.00202 | 0.00235 | 0.00605 | 0.01831 | 0.01831 |
| 4 | 136.926 | 159.866 | 131.276 | 0.00293 | 0.00293 | 0.00118 | 0.00153 | 0.00355 | 0.01259 | 0.01259 |
| 5 | 151.989 | 157.339 | 132.857 | 0.00174 | 0.00174 | 0.00075 | 0.00096 | 0.00225 | 0.01024 | 0.01024 |
| 6 | 121.345 | 139.644 | 98.25 | 0.00684 | 0.00684 | 0.00388 | 0.00332 | 0.01164 | 0.02534 | 0.02534 |
| 7 | 165.4 | 210.5 | 75.3 | 0.0078 | 0.0078 | 0.0039 | 0.0046 | 0.012 | 0.068 | 0.068 |
| 8 | 113.2 | 125.8 | 102.5 | 0.0049 | 0.0049 | 0.0024 | 0.003 | 0.0072 | 0.057 | 0.057 |
| 9 | 149.8 | 162.5 | 142.3 | 0.004 | 0.004 | 0.002 | 0.0024 | 0.0061 | 0.018 | 0.018 |
| 10 | 152.3 | 158.9 | 133.4 | 0.0018 | 0.0018 | 0.0008 | 0.001 | 0.0023 | 0.01 | 0.01 |
| 11 | 120.5 | 140.2 | 97.8 | 0.0069 | 0.0069 | 0.0039 | 0.0033 | 0.0117 | 0.025 | 0.025 |
| 12 | 160.2 | 169.0 | 145.2 | 0.0034 | 0.0034 | 0.0018 | 0.0018 | 0.0054 | 0.034 | 0.034 |
| 13 | 242.852 | 255.034 | 227.911 | 0.00225 | 0.00225 | 0.00117 | 0.00139 | 0.0035 | 0.01494 | 0.01494 |
| 14 | 139.224 | 586.567 | 66.157 | 0.03011 | 0.03011 | 0.01854 | 0.01628 | 0.05563 | 0.09419 | 0.09419 |
| 15 | 186.163 | 197.724 | 177.584 | 0.00298 | 0.00298 | 0.00165 | 0.00175 | 0.00496 | 0.01495 | 0.01495 |
| 16 | 116.879 | 131.897 | 108.153 | 0.00788 | 0.00788 | 0.00334 | 0.00493 | 0.01003 | 0.02645 | 0.02645 |

## B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 186.695 | 211.961 | 149.442 | 0.01872 | 0.01872 | 0.01075 | 0.01154 | 0.03225 | 0.0583 | 0.0583 |
| 2 | 110.793 | 128.101 | 107.316 | 0.00494 | 0.00494 | 0.0026 | 0.00283 | 0.0078 | 0.02442 | 0.02442 |
| 3 | 117.87 | 127.349 | 95.654 | 0.00647 | 0.00647 | 0.00356 | 0.003 | 0.01067 | 0.03087 | 0.03087 |
| 4 | 116.848 | 217.552 | 99.503 | 0.00531 | 0.00531 | 0.0026 | 0.00346 | 0.0078 | 0.01795 | 0.01795 |
| 5 | 229.401 | 252.221 | 221.156 | 0.00205 | 0.00205 | 0.00114 | 0.00113 | 0.00342 | 0.01457 | 0.01457 |
| 6 | 176.824 | 215.293 | 83.961 | 0.0046 | 0.0046 | 0.00209 | 0.00221 | 0.00628 | 0.01169 | 0.01169 |
| 7 | 180.45 | 210.75 | 150.3 | 0.0185 | 0.0185 | 0.0108 | 0.0116 | 0.032 | 0.058 | 0.058 |
| 8 | 115.6 | 125.8 | 105.5 | 0.005 | 0.005 | 0.0027 | 0.0029 | 0.0078 | 0.0245 | 0.0245 |
| 9 | 118.2 | 128.5 | 96.0 | 0.0065 | 0.0065 | 0.0036 | 0.003 | 0.0107 | 0.031 | 0.031 |
| 10 | 115.8 | 218.0 | 100.0 | 0.0053 | 0.0053 | 0.0026 | 0.0035 | 0.0078 | 0.018 | 0.018 |
| 11 | 230.0 | 255.0 | 220.0 | 0.0021 | 0.0021 | 0.0011 | 0.0011 | 0.0034 | 0.0146 | 0.0146 |
| 12 | 177.0 | 215.0 | 85.0 | 0.0046 | 0.0046 | 0.0021 | 0.0022 | 0.0063 | 0.0117 | 0.0117 |
| 13 | 208.519 | 220.315 | 199.02 | 0.00609 | 0.00609 | 0.00368 | 0.00339 | 0.01105 | 0.01761 | 0.01761 |
| 14 | 173.898 | 211.35 | 74.677 | 0.00448 | 0.00448 | 0.00237 | 0.00254 | 0.0071 | 0.06727 | 0.06727 |
| 15 | 116.014 | 141.781 | 110.655 | 0.01284 | 0.01284 | 0.00655 | 0.00908 | 0.01966 | 0.06425 | 0.06425 |
| 16 | 98.804 | 102.305 | 87.804 | 0.00432 | 0.00432 | 0.00247 | 0.00258 | 0.00742 | 0.02223 | 0.02223 |

## B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 112.547 | 133.374 | 105.715 | 0.00355 | 0.00355 | 0.00166 | 0.0019 | 0.00499 | 0.01358 | 0.01358 |
| 2 | 173.917 | 192.735 | 86.18 | 0.00476 | 0.00476 | 0.00221 | 0.00258 | 0.00663 | 0.04192 | 0.04192 |
| 3 | 197.076 | 206.896 | 192.055 | 0.00289 | 0.00289 | 0.00166 | 0.00168 | 0.00498 | 0.01098 | 0.01098 |
| 4 | 174.688 | 240.005 | 74.287 | 0.0136 | 0.0136 | 0.00624 | 0.00564 | 0.01873 | 0.02308 | 0.02308 |
| 5 | 110.568 | 125.394 | 106.821 | 0.00462 | 0.00462 | 0.00226 | 0.0028 | 0.00677 | 0.02199 | 0.02199 |
| 6 | 187.733 | 202.324 | 173.015 | 0.00316 | 0.00316 | 0.00168 | 0.00182 | 0.00504 | 0.01663 | 0.01663 |
| 7 | 125.34 | 210.45 | 85.67 | 0.0032 | 0.0032 | 0.0015 | 0.0018 | 0.0045 | 0.012 | 0.012 |
| 8 | 138.55 | 195.23 | 92.45 | 0.0041 | 0.0041 | 0.0021 | 0.0024 | 0.0058 | 0.015 | 0.015 |
| 9 | 115.78 | 180.67 | 78.34 | 0.0029 | 0.0029 | 0.0014 | 0.0016 | 0.0042 | 0.011 | 0.011 |
| 10 | 142.89 | 220.12 | 88.76 | 0.0038 | 0.0038 | 0.0019 | 0.0022 | 0.005 | 0.013 | 0.013 |
| 11 | 130.45 | 205.67 | 90.12 | 0.0043 | 0.0043 | 0.0022 | 0.0025 | 0.0062 | 0.014 | 0.014 |
| 12 | 125.89 | 198.45 | 85.34 | 0.0035 | 0.0035 | 0.0017 | 0.0019 | 0.0048 | 0.0125 | 0.0125 |
| 13 | 177.876 | 192.921 | 168.013 | 0.00411 | 0.00411 | 0.00233 | 0.00241 | 0.007 | 0.02126 | 0.02126 |
| 14 | 119.992 | 157.302 | 74.997 | 0.00784 | 0.00784 | 0.0037 | 0.00554 | 0.01109 | 0.04374 | 0.04374 |
| 15 | 116.556 | 592.03 | 86.228 | 0.00496 | 0.00496 | 0.00254 | 0.00263 | 0.00762 | 0.0166 | 0.0166 |
| 16 | 148.272 | 164.989 | 142.299 | 0.00459 | 0.00459 | 0.0025 | 0.00256 | 0.0075 | 0.0419 | 0.0419 |

## B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 119.056 | 125.213 | 86.795 | 0.00346 | 0.00346 | 0.00169 | 0.0017 | 0.00508 | 0.01201 | 0.01201 |
| 2 | 180.198 | 201.249 | 175.456 | 0.00284 | 0.00284 | 0.00153 | 0.00166 | 0.00459 | 0.01444 | 0.01444 |
| 3 | 114.238 | 124.393 | 77.022 | 0.00581 | 0.00581 | 0.00299 | 0.00316 | 0.00896 | 0.04009 | 0.04009 |
| 4 | 119.1 | 128.442 | 88.833 | 0.00692 | 0.00692 | 0.00393 | 0.003 | 0.01179 | 0.02682 | 0.02682 |
| 5 | 209.516 | 253.017 | 89.488 | 0.00564 | 0.00564 | 0.00331 | 0.00292 | 0.00994 | 0.02751 | 0.02751 |
| 6 | 107.332 | 113.84 | 104.315 | 0.0029 | 0.0029 | 0.00144 | 0.00182 | 0.00431 | 0.01567 | 0.01567 |
| 7 | 130.45 | 210.67 | 75.89 | 0.0032 | 0.0032 | 0.0016 | 0.0018 | 0.0055 | 0.0132 | 0.0132 |
| 8 | 145.78 | 195.34 | 80.56 | 0.0028 | 0.0028 | 0.0014 | 0.0016 | 0.0049 | 0.0125 | 0.0125 |
| 9 | 125.89 | 185.23 | 70.12 | 0.0037 | 0.0037 | 0.0018 | 0.002 | 0.006 | 0.015 | 0.015 |
| 10 | 138.56 | 220.45 | 78.34 | 0.003 | 0.003 | 0.0015 | 0.0017 | 0.0052 | 0.0138 | 0.0138 |
| 11 | 132.34 | 198.76 | 73.45 | 0.0035 | 0.0035 | 0.0017 | 0.0019 | 0.0058 | 0.0145 | 0.0145 |
| 12 | 127.89 | 188.45 | 69.78 | 0.0034 | 0.0034 | 0.0016 | 0.0018 | 0.0054 | 0.014 | 0.014 |
| 13 | 100.96 | 110.019 | 95.628 | 0.00606 | 0.00606 | 0.00351 | 0.00348 | 0.01053 | 0.02427 | 0.02427 |
| 14 | 240.301 | 245.135 | 219.783 | 0.00517 | 0.00517 | 0.00316 | 0.00283 | 0.00949 | 0.02018 | 0.02018 |
| 15 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 16 | 152.125 | 161.469 | 76.596 | 0.00382 | 0.00382 | 0.00191 | 0.00226 | 0.00574 | 0.05925 | 0.05925 |

## B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 126.344 | 134.231 | 112.773 | 0.00448 | 0.00448 | 0.00131 | 0.00169 | 0.00393 | 0.02033 | 0.02033 |
| 2 | 119.056 | 125.213 | 86.795 | 0.00346 | 0.00346 | 0.00169 | 0.0017 | 0.00508 | 0.01201 | 0.01201 |
| 3 | 208.519 | 220.315 | 199.02 | 0.00609 | 0.00609 | 0.00368 | 0.00339 | 0.01105 | 0.01761 | 0.01761 |
| 4 | 217.116 | 233.481 | 93.978 | 0.00404 | 0.00404 | 0.00127 | 0.00128 | 0.00381 | 0.01299 | 0.01299 |
| 5 | 206.327 | 226.355 | 92.02 | 0.00495 | 0.00495 | 0.00302 | 0.00246 | 0.00905 | 0.02105 | 0.02105 |
| 6 | 150.44 | 163.441 | 144.736 | 0.00396 | 0.00396 | 0.00206 | 0.00233 | 0.00619 | 0.02551 | 0.02551 |
| 7 | 124.56 | 132.78 | 111.45 | 0.0042 | 0.0042 | 0.0013 | 0.0017 | 0.0038 | 0.0201 | 0.0201 |
| 8 | 118.75 | 125.89 | 85.67 | 0.0035 | 0.0035 | 0.0017 | 0.0017 | 0.005 | 0.0118 | 0.0118 |
| 9 | 210.45 | 219.87 | 198.3 | 0.0061 | 0.0061 | 0.0037 | 0.0034 | 0.011 | 0.0174 | 0.0174 |
| 10 | 216.78 | 234.12 | 94.5 | 0.004 | 0.004 | 0.0013 | 0.0013 | 0.0038 | 0.013 | 0.013 |
| 11 | 205.34 | 225.45 | 92.1 | 0.005 | 0.005 | 0.003 | 0.0025 | 0.009 | 0.021 | 0.021 |
| 12 | 149.8 | 164.2 | 144.0 | 0.0039 | 0.0039 | 0.002 | 0.0023 | 0.0062 | 0.0255 | 0.0255 |
| 13 | 129.336 | 139.867 | 118.604 | 0.0049 | 0.0049 | 0.00165 | 0.00183 | 0.00495 | 0.02498 | 0.02498 |
| 14 | 120.256 | 125.306 | 104.773 | 0.00407 | 0.00407 | 0.00224 | 0.00205 | 0.00671 | 0.01516 | 0.01516 |
| 15 | 116.15 | 131.731 | 109.815 | 0.00381 | 0.00381 | 0.00181 | 0.00232 | 0.00542 | 0.03026 | 0.03026 |
| 16 | 117.87 | 127.349 | 95.654 | 0.00647 | 0.00647 | 0.00356 | 0.003 | 0.01067 | 0.03087 | 0.03087 |

## B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 182.018 | 197.173 | 79.187 | 0.00842 | 0.00842 | 0.00506 | 0.00449 | 0.01517 | 0.02503 | 0.02503 |
| 2 | 114.563 | 119.167 | 86.647 | 0.00327 | 0.00327 | 0.00146 | 0.00184 | 0.00439 | 0.01185 | 0.01185 |
| 3 | 177.876 | 192.921 | 168.013 | 0.00411 | 0.00411 | 0.00233 | 0.00241 | 0.007 | 0.02126 | 0.02126 |
| 4 | 208.519 | 220.315 | 199.02 | 0.00609 | 0.00609 | 0.00368 | 0.00339 | 0.01105 | 0.01761 | 0.01761 |
| 5 | 128.001 | 138.052 | 122.08 | 0.00436 | 0.00436 | 0.00137 | 0.00166 | 0.00411 | 0.02297 | 0.02297 |
| 6 | 210.141 | 232.706 | 185.258 | 0.00534 | 0.00534 | 0.00321 | 0.0028 | 0.00964 | 0.0168 | 0.0168 |
| 7 | 180.45 | 195.32 | 80.12 | 0.0087 | 0.0086 | 0.0052 | 0.0045 | 0.0153 | 0.0251 | 0.025 |
| 8 | 115.67 | 120.45 | 85.23 | 0.0033 | 0.0032 | 0.0015 | 0.0018 | 0.0044 | 0.0119 | 0.0118 |
| 9 | 178.9 | 193.8 | 168.5 | 0.0041 | 0.0041 | 0.0023 | 0.0024 | 0.007 | 0.0213 | 0.0212 |
| 10 | 209.5 | 220.8 | 199.1 | 0.0061 | 0.0061 | 0.0037 | 0.0034 | 0.0111 | 0.0176 | 0.0176 |
| 11 | 128.5 | 138.1 | 122.2 | 0.0044 | 0.0044 | 0.0014 | 0.0017 | 0.0041 | 0.0229 | 0.0229 |
| 12 | 210.2 | 232.8 | 185.3 | 0.0053 | 0.0053 | 0.0032 | 0.0028 | 0.0096 | 0.0168 | 0.0168 |
| 13 | 100.96 | 110.019 | 95.628 | 0.00606 | 0.00606 | 0.00351 | 0.00348 | 0.01053 | 0.02427 | 0.02427 |
| 14 | 163.656 | 200.841 | 76.779 | 0.00742 | 0.00742 | 0.0038 | 0.0039 | 0.0114 | 0.01659 | 0.01659 |
| 15 | 148.09 | 162.824 | 67.343 | 0.00762 | 0.00762 | 0.00467 | 0.00354 | 0.014 | 0.05428 | 0.05428 |
| 16 | 153.848 | 165.738 | 65.782 | 0.0084 | 0.0084 | 0.00428 | 0.0045 | 0.01285 | 0.0381 | 0.0381 |

## B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 162.568 | 198.346 | 77.63 | 0.00502 | 0.00502 | 0.0028 | 0.00253 | 0.00841 | 0.01791 | 0.01791 |
| 2 | 110.417 | 131.067 | 103.37 | 0.00784 | 0.00784 | 0.00352 | 0.00514 | 0.01056 | 0.03715 | 0.03715 |
| 3 | 110.568 | 125.394 | 106.821 | 0.00462 | 0.00462 | 0.00226 | 0.0028 | 0.00677 | 0.02199 | 0.02199 |
| 4 | 122.188 | 128.611 | 115.765 | 0.00524 | 0.00524 | 0.00169 | 0.00203 | 0.00507 | 0.01613 | 0.01613 |
| 5 | 120.08 | 139.71 | 111.208 | 0.00405 | 0.00405 | 0.0018 | 0.0022 | 0.0054 | 0.01706 | 0.01706 |
| 6 | 180.198 | 201.249 | 175.456 | 0.00284 | 0.00284 | 0.00153 | 0.00166 | 0.00459 | 0.01444 | 0.01444 |
| 7 | 160.45 | 195.32 | 78.12 | 0.0058 | 0.0058 | 0.0031 | 0.0027 | 0.0092 | 0.0185 | 0.0185 |
| 8 | 112.3 | 130.5 | 102.8 | 0.0072 | 0.0072 | 0.0036 | 0.0053 | 0.011 | 0.036 | 0.036 |
| 9 | 109.8 | 124.5 | 107.2 | 0.0048 | 0.0048 | 0.0024 | 0.0029 | 0.0065 | 0.0225 | 0.0225 |
| 10 | 121.5 | 128.7 | 116.2 | 0.0054 | 0.0054 | 0.0017 | 0.0021 | 0.0053 | 0.0165 | 0.0165 |
| 11 | 180.5 | 200.0 | 175.0 | 0.0029 | 0.0029 | 0.0015 | 0.0016 | 0.0046 | 0.0144 | 0.0144 |
| 12 | 202.0 | 210.5 | 197.3 | 0.0018 | 0.0018 | 0.0009 | 0.0011 | 0.0028 | 0.0095 | 0.0095 |
| 13 | 126.144 | 154.284 | 97.543 | 0.00975 | 0.00975 | 0.00593 | 0.00454 | 0.01778 | 0.02852 | 0.02852 |
| 14 | 223.361 | 263.872 | 87.638 | 0.00352 | 0.00352 | 0.00169 | 0.00188 | 0.00506 | 0.02536 | 0.02536 |
| 15 | 112.014 | 588.518 | 107.024 | 0.00533 | 0.00533 | 0.00268 | 0.00329 | 0.00805 | 0.02448 | 0.02448 |
| 16 | 107.332 | 113.84 | 104.315 | 0.0029 | 0.0029 | 0.00144 | 0.00182 | 0.00431 | 0.01567 | 0.01567 |

## B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

| # | MDVP:Fo | MDVP:Fhi | MDVP:Flo | MDVP:Jitter | MDVP:Jitter.1 | MDVP:RAP | MDVP:PPQ | Jitter:DDP | MDVP:Shimmer | MDVP:Shimmer.1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 187.733 | 202.324 | 173.015 | 0.00316 | 0.00316 | 0.00168 | 0.00182 | 0.00504 | 0.01663 | 0.01663 |
| 2 | 102.273 | 142.83 | 85.902 | 0.00907 | 0.00907 | 0.00493 | 0.00461 | 0.0148 | 0.02814 | 0.02814 |
| 3 | 163.656 | 200.841 | 76.779 | 0.00742 | 0.00742 | 0.0038 | 0.0039 | 0.0114 | 0.01659 | 0.01659 |
| 4 | 237.226 | 247.326 | 225.227 | 0.00298 | 0.00298 | 0.00169 | 0.00182 | 0.00507 | 0.01752 | 0.01752 |
| 5 | 112.014 | 588.518 | 107.024 | 0.00533 | 0.00533 | 0.00268 | 0.00329 | 0.00805 | 0.02448 | 0.02448 |
| 6 | 112.15 | 131.669 | 97.527 | 0.00519 | 0.00519 | 0.00291 | 0.00284 | 0.00873 | 0.01756 | 0.01756 |
| 7 | 180.45 | 210.12 | 165.78 | 0.0032 | 0.0032 | 0.0017 | 0.0019 | 0.0051 | 0.0172 | 0.0172 |
| 8 | 150.3 | 190.5 | 140.2 | 0.0041 | 0.0041 | 0.0022 | 0.0024 | 0.0063 | 0.0205 | 0.0205 |
| 9 | 165.2 | 205.7 | 155.4 | 0.0038 | 0.0038 | 0.002 | 0.0022 | 0.0058 | 0.0189 | 0.0189 |
| 10 | 200.1 | 240.3 | 180.5 | 0.0029 | 0.0029 | 0.0016 | 0.0018 | 0.0049 | 0.0158 | 0.0158 |
| 11 | 120.5 | 160.8 | 110.2 | 0.0052 | 0.0052 | 0.0028 | 0.003 | 0.008 | 0.022 | 0.022 |
| 12 | 175.4 | 215.6 | 160.7 | 0.0035 | 0.0035 | 0.0019 | 0.0021 | 0.0055 | 0.0178 | 0.0178 |
| 13 | 128.001 | 138.052 | 122.08 | 0.00436 | 0.00436 | 0.00137 | 0.00166 | 0.00411 | 0.02297 | 0.02297 |
| 14 | 155.078 | 163.736 | 144.148 | 0.00168 | 0.00168 | 0.00068 | 0.00092 | 0.00204 | 0.01064 | 0.01064 |
| 15 | 157.821 | 172.975 | 68.401 | 0.00358 | 0.00358 | 0.00196 | 0.00196 | 0.00587 | 0.03716 | 0.03716 |
| 16 | 129.336 | 139.867 | 118.604 | 0.0049 | 0.0049 | 0.00165 | 0.00183 | 0.00495 | 0.02498 | 0.02498 |